PN-Junction Diode

Outline

- The *pn* Junction
  - PN-Junction Diode
- Analysis of diode circuits
- Applications of diode circuits
  - Rectification
    - Half wave Rectifiers
    - Full wave Rectifiers

1.3 *pn* Junction

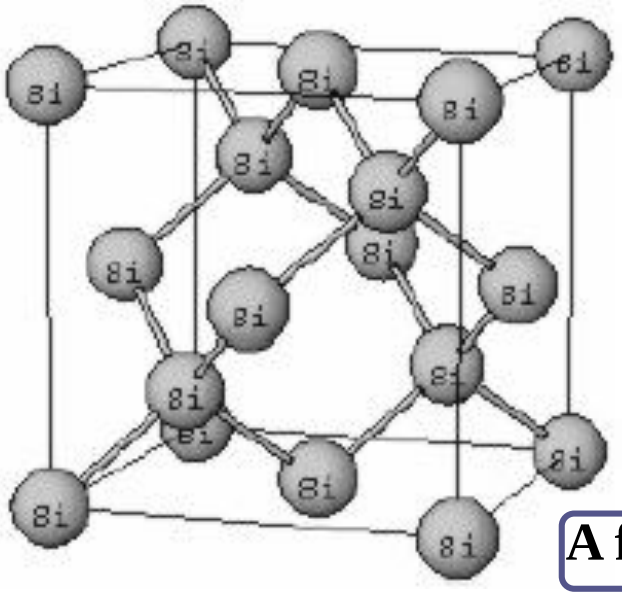
- The *pn* junction under open-circuit condition
- I-V characteristic of *pn* junction
  - Terminal characteristic of junction diode.
  - Physical operation of diode.
- Junction capacitance

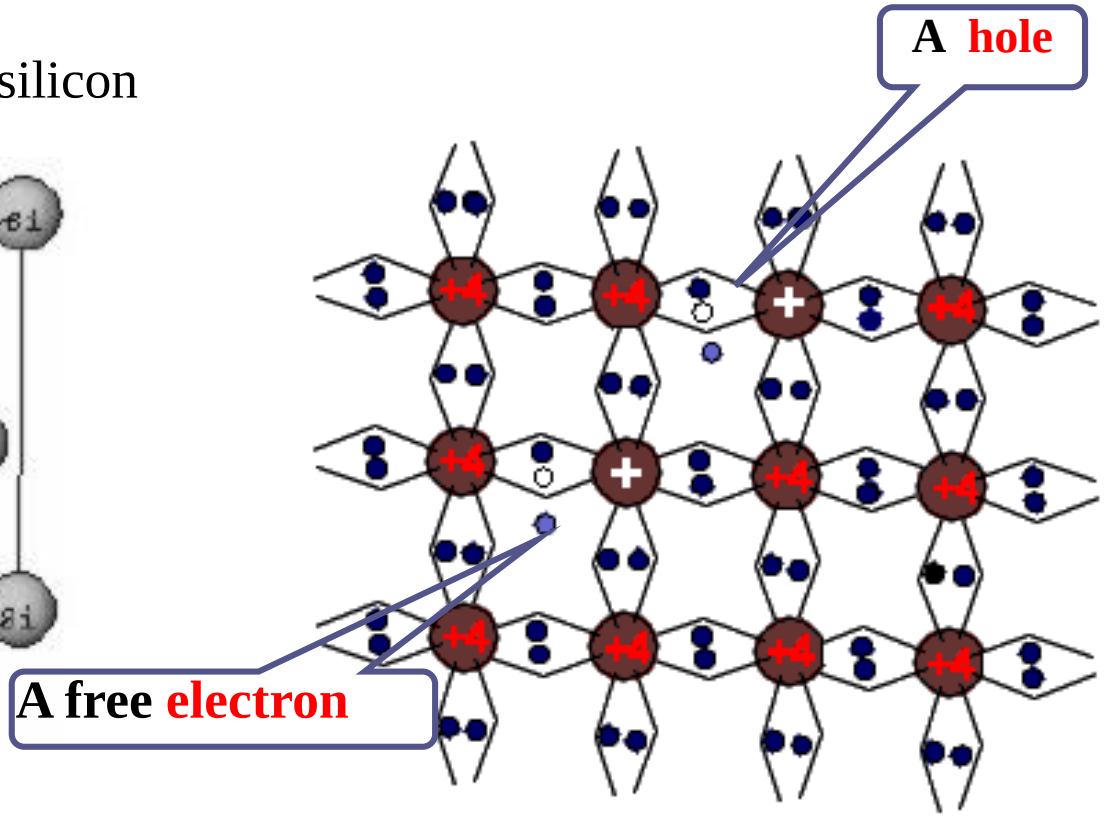
Previous Lecture

- Semiconductor
  - Intrinsic
  - Doping
  - Extrinsic
    - N-type
    - P-type

Intrinsic (pure) Semiconductors

- Intrinsic(pure) silicon





• An electron-hole pair is created when an electron get excited by thermal or light energy;

• Recombination occurs when an electron loses energy and falls back into a hole.

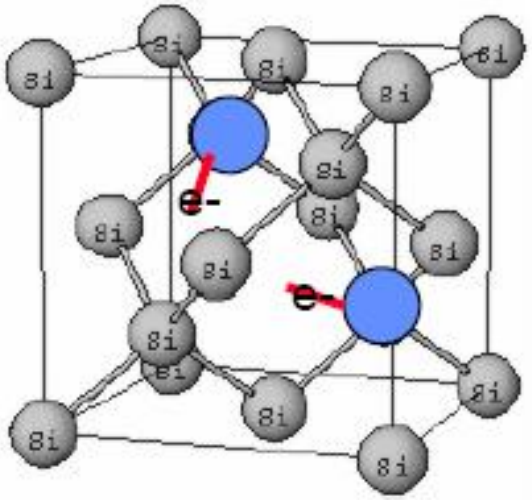
Intrinsic (pure) Semiconductors

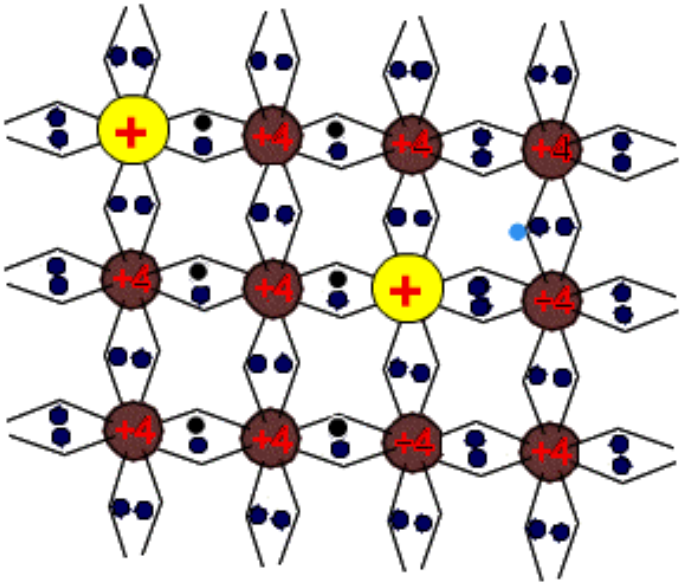
- Holes also conduct current. In reality, it's the movement of all the other electrons. The hole allows this motion.
- Holes have positive charge.
- Current flows in the same direction as the holes move.

- Both electrons and holes carry current-- **carriers**.
- In intrinsic semiconductors the electron and hole concentrations are equal because carriers are created in pairs
- The intrinsic concentration depends exponentially on temperature.
- At room temp (300K), the intrinsic carrier concentration of silicon is:

$$n_i = 1.5 \times 10^{10} / cm^3$$

Phosphorus Doping (N-type)





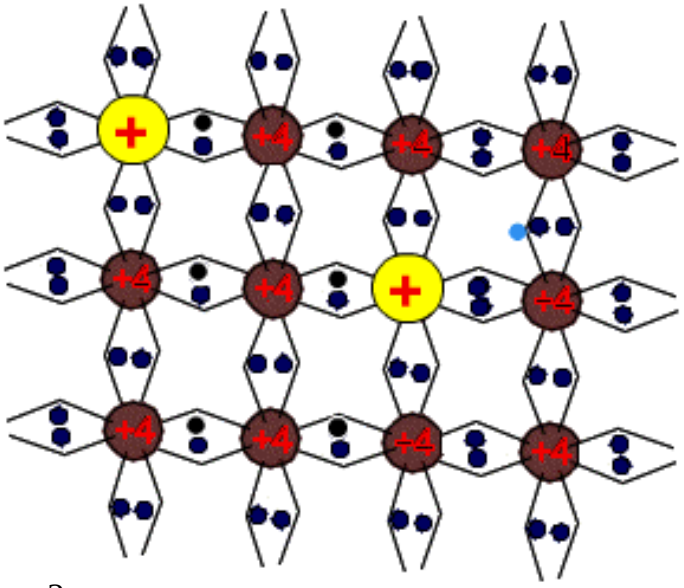
- Phosphorus has 5 valence electrons.
- P atoms will sit in the location of a Si atom in the lattice, to avoid breaking symmetry, but each will have an extra electron that does not bond in the same way. And these extra electrons are easier to excite (and can move around more easily)
- These electrons depends on the amounts of the two materials.

Phosphorus Doping (N-type)

Electrons---Majority carrier.

Holes---Minority carrier

Phosphorus---**Donor materials.**



• In equilibrium, $pn = p_i n_i = p_i^2 = n_i^2$

• At room temp (300K), if $1/10^{10}$ donors are added to the intrinsic silicon, then the electron carrier concentration is about 10^{13}cm^{-3}; the hole carrier concentration is about 10^{6}cm^{-3}.

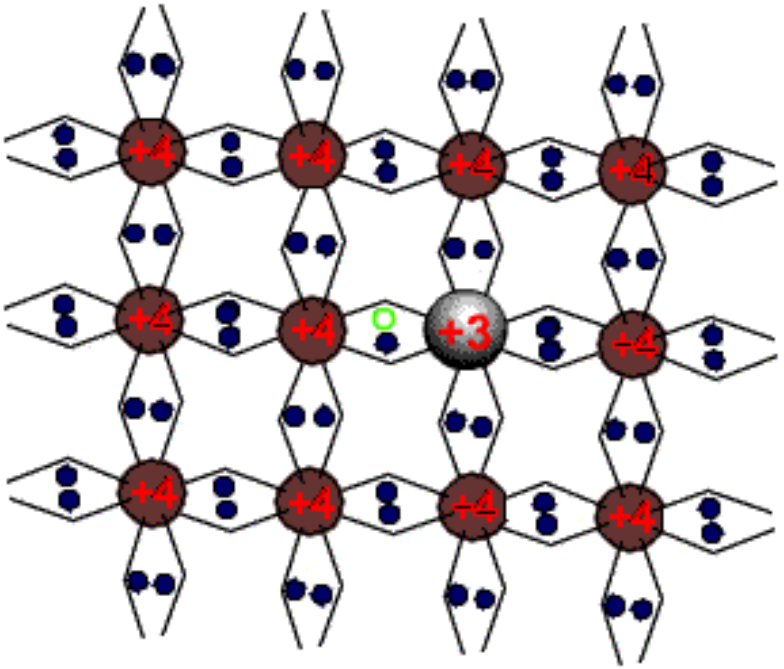
Phosphorus $\rho \approx 89.3\Omega \cdot cm$; Intrinsic silicon $\rho \approx 2.14\times 10^{5}\Omega \cdot cm$

Boron Doping (P-type)

Holes---Majority carrier;

Electrons---Minority carrier

Boron---**acceptor materials.**

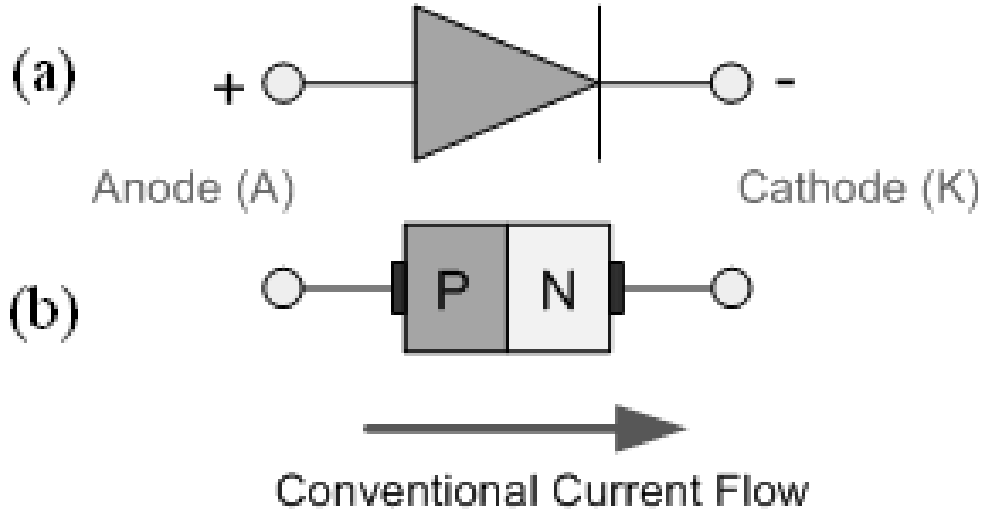


• Boron has 3 valence electrons.

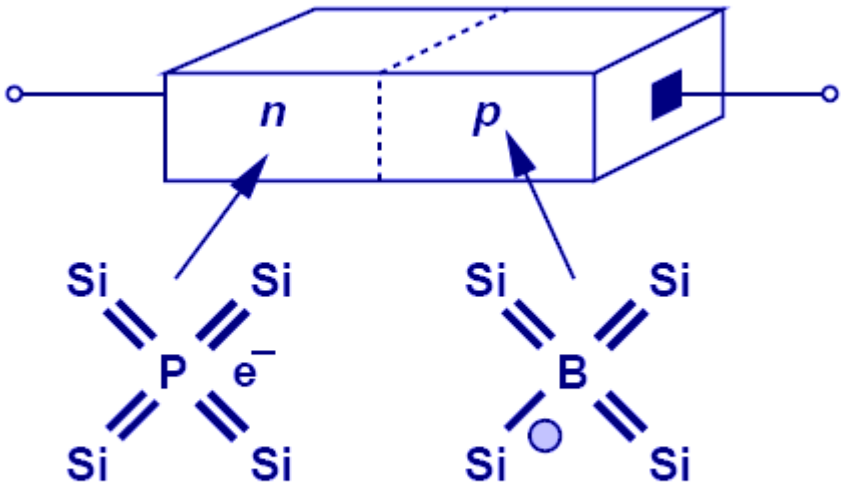
• B will sit at a lattice site, but the adjacent Si atoms lack an electron to fill its shell. This creates a hole.

PN Junction

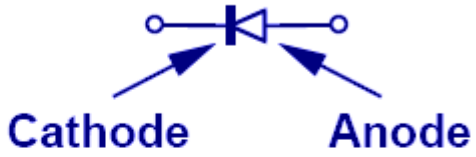
- A p-n junction is the junction between an n-type semiconductor and a p-type semiconductor
- It is fabricated by introducing donor impurities into one side of an intrinsic semiconductor crystal and acceptor impurities into the other side.
- It is also called a semiconductor diode
- The figure below shows the schematic diagram (a) and block diagram (b) of the PN-junction



PN Junction



(a)

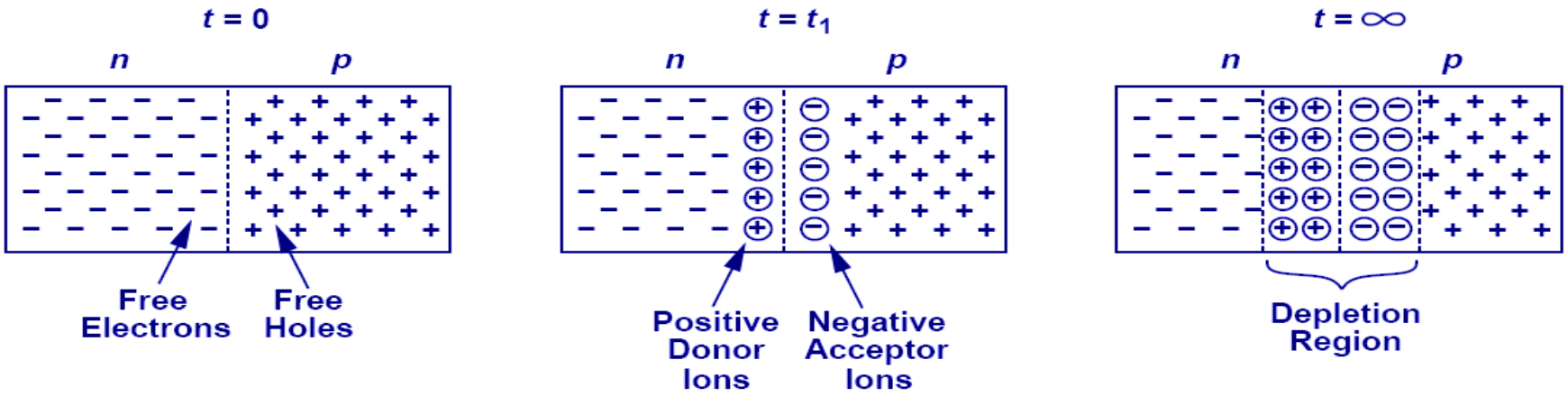


(b)

- When N-type and P-type dopants are introduced side-by-side in a semiconductor, a PN junction or a diode is formed.

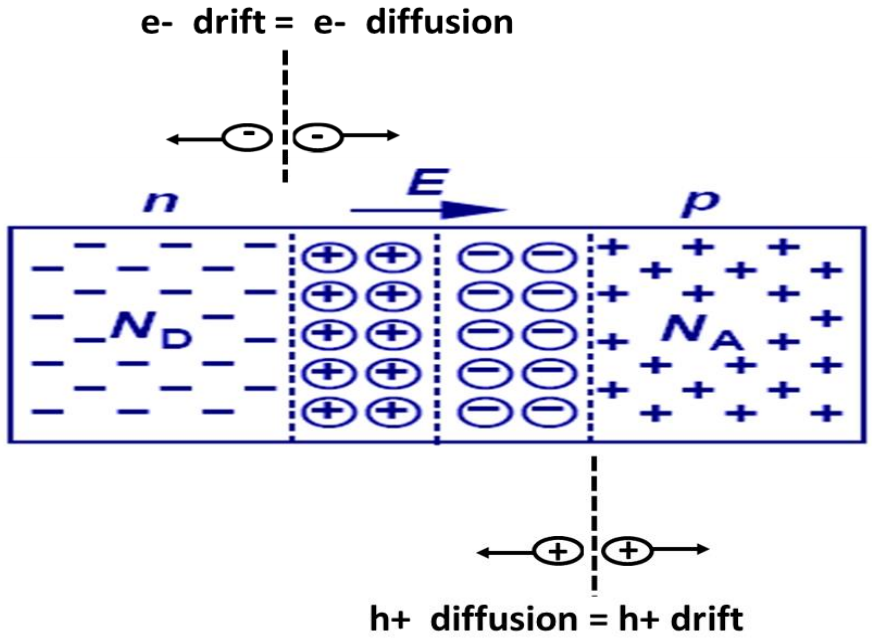
PN Junction

- The figure below illustrates the cross-section of a p-n junction
- Assume the p-n junction is initially made (t=0); A diffusion current consisting of both holes and free electrons will flow across the junction
- Holes diffuse out of the p-type side and into the n-type side and free electrons diffuse out of the n-type side and into the p-type-side
- The p-side becomes negatively charged and the n-side becomes positively charged (t = t1), caused by the ionized acceptor atoms and ionized donor atoms on the p-side and n-side respectively



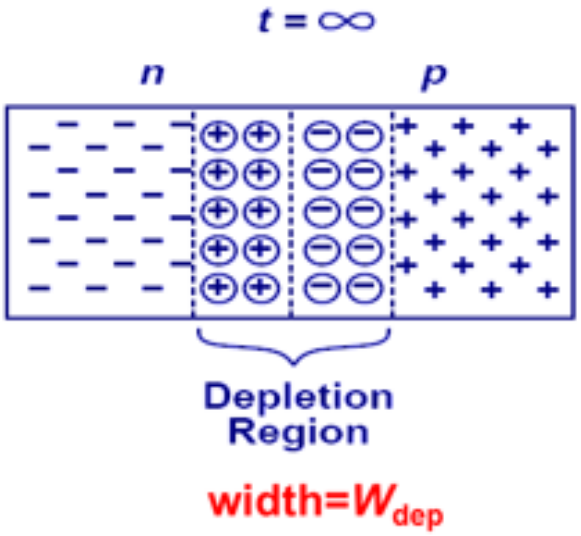
PN Junction in Equilibrium

- The charges on either side of the p-n junction cause an electric field to build up across the junction which is directed from the n-type side to the p-type side
- The electric field opposes the diffusion of holes and electrons across the junction.
- Equilibrium is reached when the force exerted on the charge carriers by the electric field is equal to the diffusion force



Depletion Region

- Because no free electrons or holes can exist in the region about the junction, there are no mobile charges to neutralize the ions in the region
- The ions on the n-type side have a positive charge on them and those on the p-type side have a negative charge
- The region about the junction in which the ions exist is called depletion region.
- Other names are the space-charge region and the transition region.
- The width of the depletion region is called the depletion width (W_{dep})

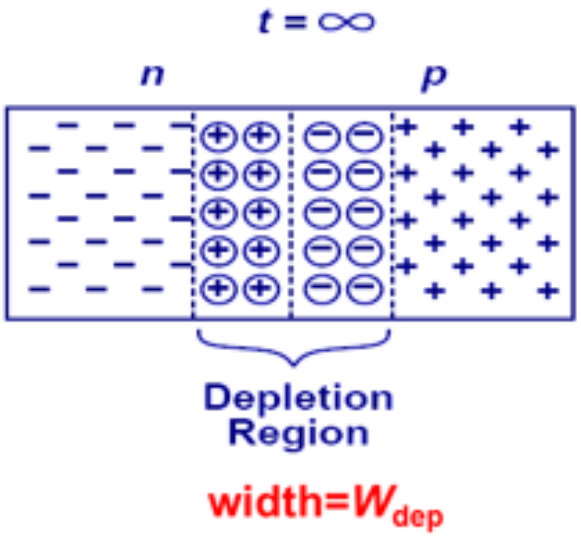


Junction Capacitance

- Junction capacitance is the capacitance associated with the depletion region
- The positive ions on the n-type side and the negative ions on the p-type side serve as the positive and negative plates respectively as in a parallel plate capacitor.
- The dielectric of the junction capacitance result from the depletion region space between the positive and negative ions
- As in a parallel plate capacitor, the junction capacitance is given by:

$$C_{dep} = \frac{\varepsilon_o \varepsilon_r A}{W_{dep}}$$

where $\varepsilon_o \varepsilon_r = \varepsilon$ is the absolute permittivity of the semiconductor
A is the cross-sectional area of the junction and
W_{dep} is the depletion width

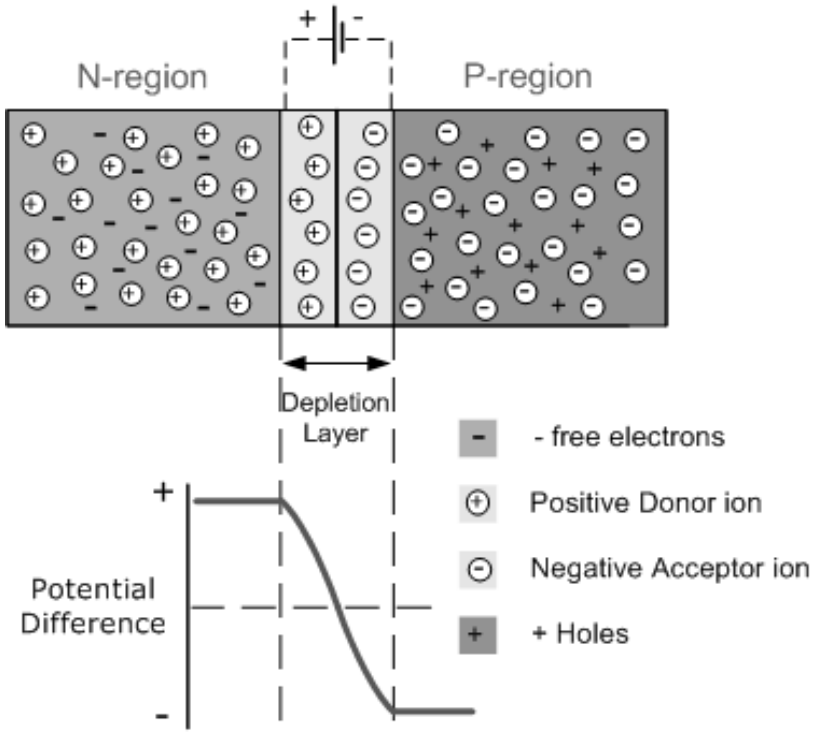


Built-in Potential

- Because there is an electric field in the depletion region of a p-n junction that is directed from the n-type side to the p-type side, it follows that there is a difference in potential or voltage across the junction
- This voltage difference is called the built-in potential or contact potential or potential barrier
- The built-in voltage V_B is given by

$$V_B = V_T In\left(\frac{N_A N_D}{n_i^2}\right)$$

where V_T is the thermal voltage

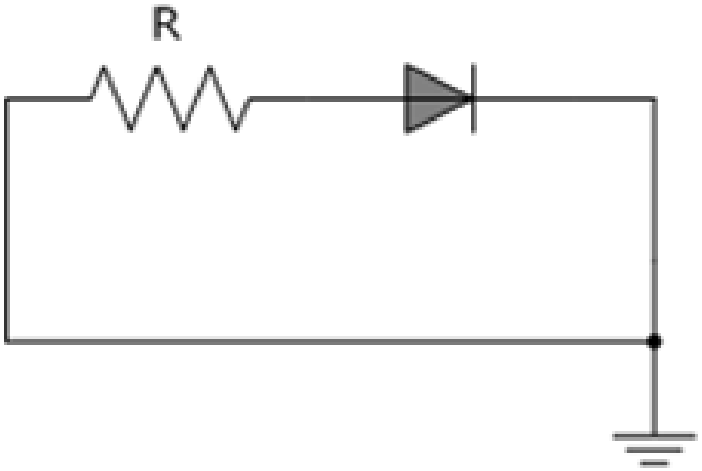


Biasing

- For semiconductors to conduct, a voltage potential is connected across it in a process called ***biasing***.
- The external voltage, based on the how the biasing is done, can increase or decrease the potential barrier.
- There are three possible biasing conditions for the standard junction diode namely: **zero biasing**, **reverse biasing** and **forward biasing**.

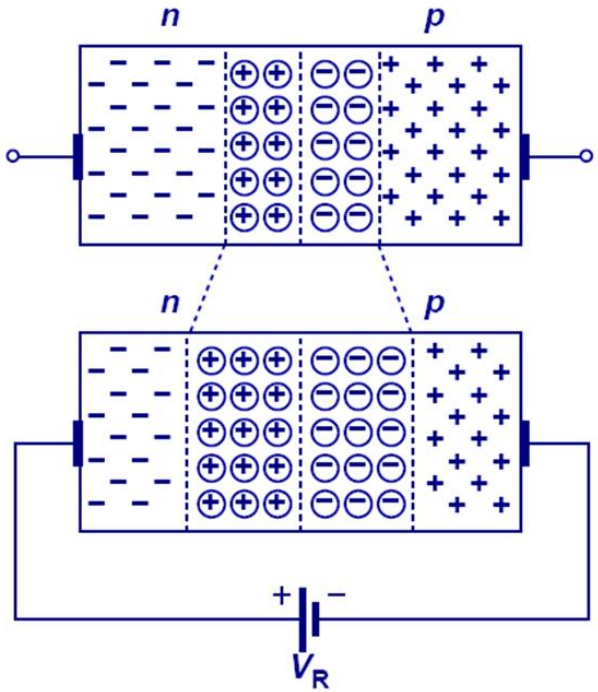
Zero Bias

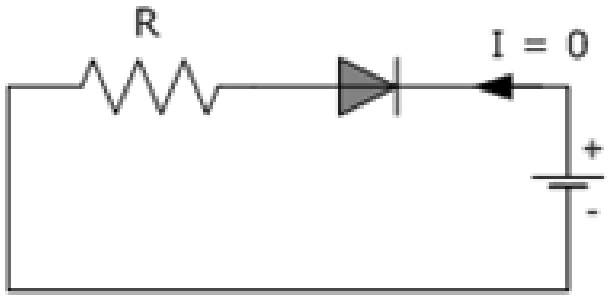
- In zero bias condition, no external voltage is applied to the PN-junction
- No current flows through the diode



Reverse Biasing

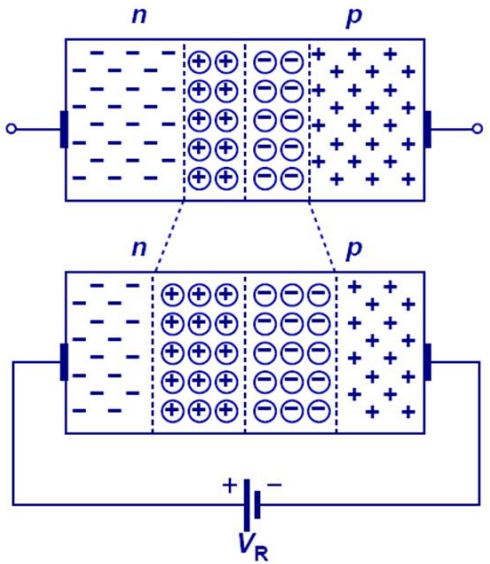
- To reverse bias a diode, a positive voltage is applied to the N-type material and a negative voltage is applied to the P-type material as shown below
- The positive voltage applied to the N-type material attracts electrons towards the positive electrode and away from the junction, while the holes in the P-type end are also attracted away from the junction towards the negative electrode.
- The net result is that, the depletion layer grows wider due to a lack of electrons and holes and presents a high impedance path.
- Consequently, a high potential barrier is created thus preventing current from flowing through the semiconductor material.





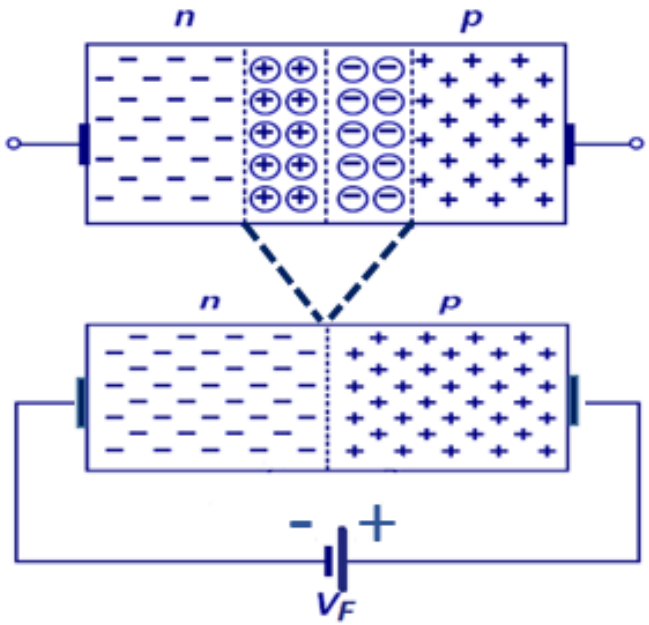
Reverse Biasing

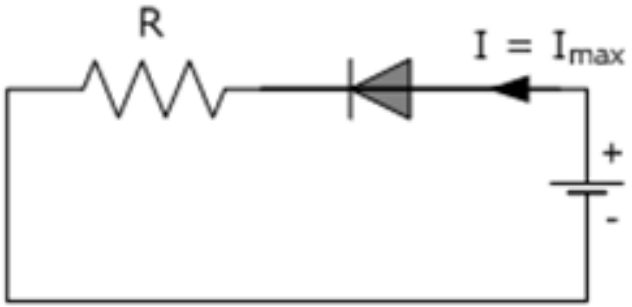
- However, a very small leakage current does flow through the junction that can be measured in microamperes, (μ *A)*.
- This leakage current is due to drift of thermally produced electron-hole pairs across the junction **i.e.** the electric field across the junction causes the electron to be swept to the n-type side and the hole to be swept to the p-type side
- If the reverse bias voltage, ***Vr*** applied to the junction is increased to a sufficiently high enough value called the **Reverse Breakdown Voltage**, it will cause the PN-junction to overheat and fail.
- This may cause the diode to become shorted and will result in maximum circuit current to flow
- This effect is called the Zener Breakdown or Avalanche Breakdown



Forward Biasing

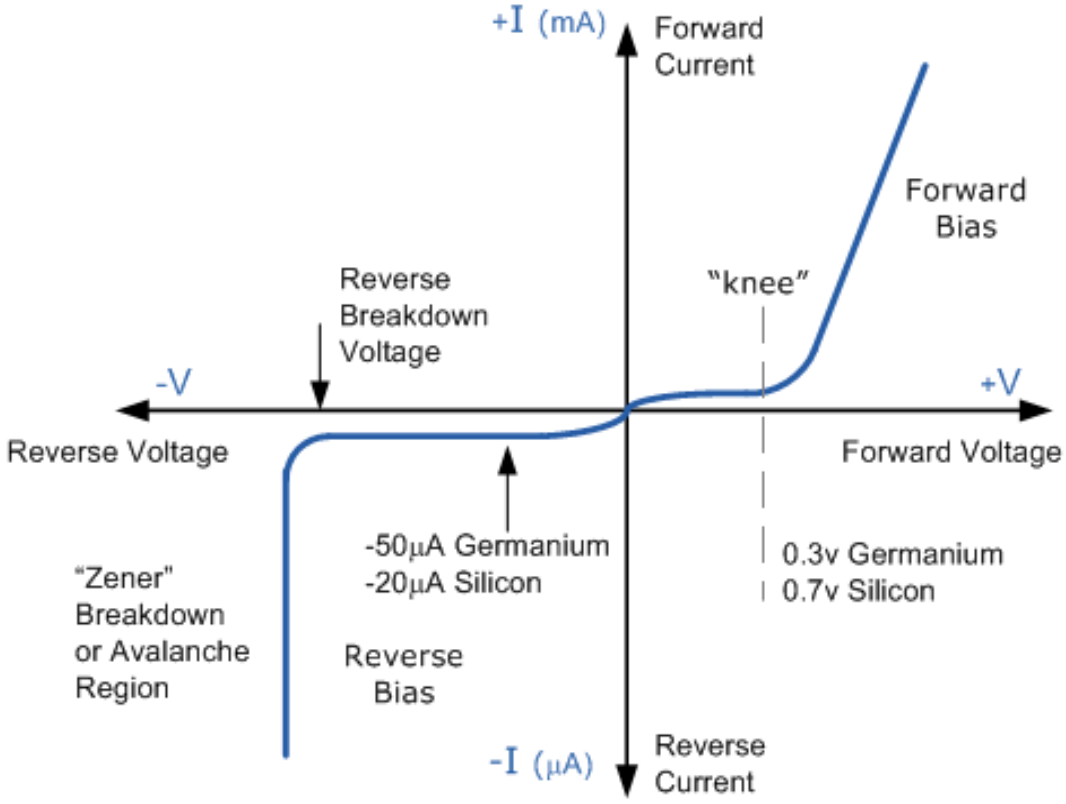
- When a diode is connected in a forward bias condition, a negative voltage is applied to the n-type material and a positive voltage is applied to the p-type material
- If this external voltage becomes greater than the value of the potential barrier, 0.7 volts for Silicon and 0.3 volts for Germanium, the potential barrier's opposition will be "subdued" and current will start to flow.
- The negative voltage repels electrons towards the junction giving them the energy to cross over and the holes are repelled in the opposite direction towards the junction by the positive voltage.
- These result in the depletion layer becoming very thin and narrow representing a low impedance path. In that event, a very small potential barrier is produced, allowing high currents to flow.
- The voltage at which the diode starts conducting is called the **knee voltage** or cut-in voltage





I-V Characteristic Curve of the P-N Diode

- The I-V Characteristic curve of the P-N diode illustrates the current (I) and voltage relationship of the diode under forward and reverse bias conditions
- It is obtained by sweeping the voltage across the diode from negative to positive whiles measuring the corresponding currents
- It is shown in the diagram below



Definition of Terms

- **Knee voltage or Cut-in Voltage:** It is the forward voltage at which the diode starts conducting.
- **Breakdown voltage**: It is the reverse voltage at which the diode (p-n junction) breaks down with sudden rise in reverse current.
- **Peak-inverse voltage (PIV):** It is the maximum reverse voltage that can be applied to a p-n junction without causing damage to the junction. If the reverse voltage across the junction exceeds its peak-inverse voltage, then the junction gets destroyed because of excessive heat. In rectification, one thing to be kept in mind is that, care should be taken that reverse voltage across the diode during negative half cycle of A.C. does not exceed the peak-inverse voltage of the diode.

It is usually safer to select a diode that has reverse breakdown voltage at least 50% greater than the expected PIV.

- **Maximum Forward current/Current handling Capacity**: It is the maximum instantaneous forward current that a p-n junction can conduct without damaging the junction.

If the forward current is more than the specified rating, then the junction gets destroyed due to overheating.